

Structural Self-Energy Expansion (SSEE-V3.6): A Minimal-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: [0009-0008-2195-7836](https://orcid.org/0009-0008-2195-7836)

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Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a dark energy framework that derives its cosmological background sector algebraically from the golden ratio φ and π , with zero fitted dimensionless parameters at the background level and no WIMP, QCD-axion, or SUSY dark-matter particle. Phenomenological extensions in the dark-matter sector (Paper 6) are currently treated as ansätze and tracked as open problems (OP-9/11, `OPEN_PROBLEMS.md`). In CMB-likelihood terms SSEE is a $k = 2$ fit: of Λ CDM's six parameters it fixes four algebraically (Ω_b, Ω_c via $\omega_c = \text{KAL}_0 \omega_b n_s$, $n_s = 1 - \varphi^{-7}$, and the derived anchor $H_0 = 3(\varphi + \pi)^2$), leaving only the primordial amplitude A_s and the optical depth τ ($k = 2$ versus $k = 6$, the count used in the Paper 3 model comparison). The dark energy equation of state $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] \simeq -0.840$ and $w_a \simeq -0.670$ are fixed by the algebraic construction and agree with the official DESI DR2 $w_0 w_a$ CDM posterior within 0.24σ (Pantheon+; $0.2\text{--}1.8\sigma$ across SN compilations)—a parameter-free postdiction, with no claim of temporal priority over DESI DR2. A Bayesian MCMC analysis under the ω_m -direct CMB anchor as prior ($H_0^{\text{prior, SSEE}} = 67.962 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the value at which Planck `plik_lite` is minimised with SSEE-fixed ω_b, ω_c ; Paper 3) yields $H_0 = 67.95^{+0.40}_{-0.40} \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.88σ from Planck, 0.04σ from the algebraic anchor) and $\Delta\text{BIC} = 5.68$ favouring SSEE over Λ CDM (5.59 over CPL). A complementary full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` + lensing + DESI DR2 + $f\sigma_8$, w_0, w_a free and H_0 under a flat prior) places the model at 1.31σ in the $w_0\text{--}w_a$ plane, where Λ CDM is excluded at 3.22σ ; under this uncompressed CMB likelihood H_0 relaxes to 65.87 ± 1.17 (1.79σ below the anchor).

The CMB-sector matter density is the standard physical observable $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$ (Planck: 0.3153 ± 0.0073), built piece-by-piece from SSEE algebra (ω_m -direct; OP-8 dissolved, no matter factor) and distinct from the dynamical $\Omega_{m,\text{dyn}} = 0.160$ that governs DESI BAO. Full-sky CLASS Boltzmann runs confirm that the *full* matter density is physically necessary: using only $\Omega_{m,\text{dyn}} = 0.160$ as the CMB matter density shifts the peak positions by $\sim 10\%$ and raises the RMS residual from 1.4% to 31.5%.

A linear Israel-Stewart perturbation analysis with $\Omega_{m,\text{cosm}} = 0.30889$ in the Poisson equation yields a growth factor $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$ (Paper 5), giving a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys — statistically identical to ΛCDM (0.73σ), resolving the growth-rate sector. A two-sector ϕ -DM extension — $\Omega_{\phi\text{DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}} = 0.14889$ active only on scales $k < k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ — reduces the S_8 lensing tension from 3.5σ (single-sector ceiling, KiDS) to 0.04σ , with $\sigma_8^{\text{eff}} = 0.747$ ($S_8^{\text{eff}} = 0.758$ from the direct two-sector CLASS run; the analytic two-sector growth factor $G_{2s} = 1.003$ is ΛCDM -consistent, the suppression to $\sigma_8^{\text{eff}} = 0.747$ arising from ϕ -DM free-streaming below k_{fs}). The same suppression reaches the $\sigma_8(R=8)$ window probed by RSD, so the two-sector $f\sigma_8$ tension rises modestly from the 0.70σ single-sector baseline to 0.93σ (still $< 1\sigma$, close to ΛCDM 0.73σ) — the deliberate cost of lowering σ_8 to resolve the lensing amplitude.

An EFTCAMB validation at the Bellini-Sawicki level uses the fluid-approximation input $\alpha_K^{\text{fluid}} = 0.4033$ ($\Delta = 0.005\%$), while the theoretical bare-field value is $\alpha_K = 0.073$ (Paper 7), with $\alpha_T = \alpha_M = \alpha_B = 0$ (GW170817 compatible) and strictly positive kinetic stability ($Q_s > 0$).

We additionally summarise three theoretical extensions of the framework. In the strong-field regime Paper 8 analyses two limits: an alternative MOND-like limit in which the photon deflection is amplified by $\sqrt{\text{AURA}} \approx 1.9995$ (numerically close to $\mathcal{M}$ at 0.03% but algebraically distinct), and the canonical two-sector limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$ in which the residual fifth-force is suppressed to $F_\phi/F_N \sim 10^{-15}$ and the observable lensing recovers GR; current SLACS/BELLS data favour the canonical limit (Paper 8; OPEN_PROBLEMS.md OP-13). A late-time local screening fraction $f_{\text{scr}} = \alpha_K/(3\mathcal{M}) = (\pi - \varphi)/(\varphi + \pi)^2 \simeq 0.067$ applied to the global background $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$ gives $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$, 0.17σ from the SH0ES distance ladder, as a zero-parameter prediction of the infrared sector (Paper 9). A UV completion $K(X) = X/\text{KAL}_0 + X^2/M^4$ with algebraic cutoff $M = 9.68 \text{ meV}$ connects the dark-energy EFT to the inflationary α -attractor (Paper 10). All model parameters are falsifiable by named forthcoming experiments.

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1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide compelling evidence for dynamical dark energy at $> 3\sigma$, with a CPL equation-of-state posterior centred near $(w_0, w_a) \approx (-0.84, -0.67)$. This remarkable coincidence motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEE-V3.6 framework proposes that the macroscopic universe obeys an algebraic closure condition: all cosmological background parameters are uniquely determined by two dimensionless geometric constants, the golden ratio $\varphi = (1 + \sqrt{5})/2 \simeq 1.6180$ and $\pi \simeq 3.1416$. No optimisation over formula space was performed: the parameter values are fixed by the algebraic construction [Almeida Vallejo, 2026k], not adjusted to the data. We make no claim of temporal priority over DESI DR2 (public from 2025 March); the agreement reported below is a parameter-free *postdiction*, and the only timestamped predictions are those reserved for unreleased data (DESI DR3, Euclid).

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration demonstrating that the full ω_m -direct matter density ($\Omega_{m,\text{CMB}} = 0.30889$) is physically necessary for reproducing Planck PR4 peak positions.
3. **Structure growth and σ_8** (§5): Israel-Stewart causal perturbation theory and a ϕ -DM two-sector extension that resolves the S_8 lensing tension via free-streaming suppression (at a modest $\sim 0.2\sigma$ cost in $f\sigma_8$, which rises to 0.93σ).
4. **EFT stability** (§6): EFTCAMB Bellini-Sawicki validation of kinetic stability (α_K) and gravitational wave compatibility ($\alpha_T = \alpha_M = \alpha_B = 0$).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic resolution of the Hubble tension, and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *postdiction* (algebraic result fixed by construction, then compared to data already public), *retrodiction* (independently derived, then compared to existing data), and *prediction* (fixed now, tested against not-yet-released data). Table 6 in §8 lists all entries explicitly.

2 Algebraic Framework

2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from φ and π :

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 && \text{(Stability Metric)} \\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 && \text{(Base Coupling Scalar)} \\
\text{KAL}_0 &= \beta + \pi \simeq 5.5214 && \text{(Structural Viscosity)} \\
P_{\text{sc}} &= \Omega + \varphi \simeq 6.3776 && \text{(Dynamical Evolution Scalar)} \\
K_v &= \varphi + \pi + \Omega \simeq 9.5192 && \text{(Structural Constraint)} \\
T_r &= 3(\varphi + \beta) \simeq 11.9935 && \text{(3D Saturation Horizon)} \\
M_v &= \varphi + \pi + K_v \simeq 14.2788 && \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is derived uniquely:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.8399, \quad w_a = -\frac{P_{\text{sc}}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \simeq -0.6699. \tag{2}$$

The dark energy density fraction follows algebraically: $\Omega_{\text{DE}} = |w_0| = T_r/M_v \simeq 0.8399$.

2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$H_0 = 3(\varphi + \pi)^2 \simeq 67.96 \text{ km s}^{-1} \text{ Mpc}^{-1}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \simeq 0.96556, \tag{4}$$

$$\Omega_{m,\text{CMB}} = \omega_m/h^2 \simeq 0.30889 \quad (\omega_m\text{-direct, CMB sector}). \tag{5}$$

Planck 2018 measurements: $H_0 = 67.36 \pm 0.54$, $n_s = 0.9649 \pm 0.0042$, $\Omega_m = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020]. All three SSEE retrodictions lie within 1.5σ of the Planck values with no adjustment.

2.3 The two- Ω_m structure and CMB matter density

The framework predicts two independent matter densities. The dark energy EoS predicts $\Omega_{\text{DE}} = |w_0| = 0.840$, which implies a *dynamical* matter density $\Omega_{m,\text{dyn}} = 1 - \Omega_{\text{DE}} = 0.160$. However, the CMB angular-diameter distance is sensitive to the *total* matter density, which under the ω_m -direct reframe (OP-8 dissolved) is the standard physical observable built from SSEE algebra:

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = 0.30889, \quad \omega_c = \text{KAL}_0 \omega_b n_s, \tag{6}$$

in agreement with Planck 2018 (0.3153 ± 0.0073) at 0.88σ . There is *no* matter factor: $\Omega_{m,\text{dyn}} = 0.160$ (DESI) and ω_m (CMB) are two independent SSEE predictions. (The earlier, now superseded MIRA mapping $\mathcal{M} = (3\varphi + \pi)/4 \simeq 1.9989$ gave $\Omega_{m,\text{CMB}} = \mathcal{M} \times \Omega_{m,\text{dyn}} = 0.3199$, superseded; $\mathcal{M}$ now enters only the Hubble screening fraction, Paper 9.) The physical interpretation of this two-sector split — whether it arises from a background-level Israel-Stewart thermodynamic correction, a two-sector dark matter architecture, or both — is examined in Papers 3 and 5 [Almeida Vallejo, 2026d,f]. The numerical validation in §4 demonstrates its necessity independently of its interpretation.

3 Background Sector Validation

3.1 Bayesian MCMC setup

We compare SSEE against Λ CDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with $N_{\text{walkers}} = 100$, $N_{\text{steps}} = 25\,000$ (2.5×10^6 total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at $z = 0.295\text{--}2.330$ [Abdul-Karim et al., 2025], with the full 13×13 same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior: $H_0 = 67.36 \pm 0.54$, $\Omega_m = 0.3153 \pm 0.0073$, $\Omega_b h^2 = 0.02237 \pm 0.00015$, $\rho(H_0, \Omega_m) = -0.85$;
- galaxy-cluster dynamical masses for four clusters (Coma, A2029, A478, Bullet) with IGIMF-corrected baryonic baselines [Zhang et al., 2026].

For SSEE, H_0 is the sole free parameter; (w_0, w_a) are fixed algebraically. For Λ CDM, (H_0, Ω_m) are free. For CPL, $(H_0, \Omega_m, w_0, w_a)$ are free.

3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results (three-model run, DESI DR2 + common Planck prior, total matter density in the background geometry). N_{eff} : effective independent samples. Positive Δ favours SSEE.

Quantity	SSEE	Λ CDM	CPL
H_0 [km s ^{−1} Mpc ^{−1}]	67.62 ± 0.40	68.27 ± 0.28	67.26 ± 0.51
Free parameters k	2	3	5
N_{eff}	78 750	64 505	44 627
ΔBIC relative to SSEE	ref	+5.68	+5.59
ΔDIC relative to SSEE	ref	+4.91	+3.27
χ_r^2 clusters	0.126	0.081	—
(w_0, w_a) vs DESI DR2 $w_0 w_a$ CDM (Pantheon+)	0.24σ	0.7σ	—

With the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.30889$ in the background geometry, SSEE is favoured on every information criterion: $\Delta\text{BIC} = 5.68$ over Λ CDM and 5.59 over CPL, $\Delta\text{DIC} = 4.91$ and 3.27. The SSEE sound horizon then matches Λ CDM ($r_d = 148.2$ Mpc vs 147.8 Mpc) and the H_0 posterior sits 0.39σ from Planck, so no background-structure penalty arises. An earlier “+206” full-background penalty was an artefact of placing the cold sector $1 + w_0 = 0.160$ in $E(z)$ where the total density belongs; with the correct total density it does not occur. The ω_m -direct two- Ω_m construction (§2.3) resolves the background tension, validated in §4 below.

3.3 Galaxy-cluster mass sector

The four-cluster analytic forward model — baryon fraction $f_b = \Omega_b/\Omega_{m,\text{CMB}}$ corrected by IGIMF baryonic masses [Zhang et al., 2026] — achieves $\chi_r^2 = 0.122$ (four-cluster MCMC

subset) and $\chi_r^2 = 0.126$ (seven-cluster analytic). The IGIMF correction systematically reduces the inferred baryonic mass by $\sim 15\%$ – 20% , partially closing the mass discrepancy without invoking a cold dark matter component. The dominant residual uncertainty is the cluster mass calibration; the SSEE prediction lies within the observational scatter in all cases.

3.4 Multi-probe MCMC validation: two complementary tests

The rigidly fixed SSEE background can be interrogated in two distinct and complementary ways, which must not be conflated. (i) *Free test*: let the data choose the equation of state and ask where they land relative to the algebraic prediction — “*what do the data prefer?*” (ii) *Fixed test*: impose the algebraic values and measure the absolute goodness of fit — “*how well does the imposed model describe the data?*” Both are performed with the *full* CMB power-spectrum likelihood — not a compressed distance prior — so that no Λ CDM-like assumption is smuggled in through the compression.

Free test (what the data prefer). We run an MCMC with CAMB + Planck `plik_lite` TTTEEE + lensing + a τ prior, combined with DESI DR2 BAO and $f\sigma_8$, sampling $(H_0, \Omega_b h^2, \Omega_c h^2, n_s, \tau, w_0, w_a)$ with *flat* priors (H_0 free in $[60, 80]$; four chains, $\sim 53k$ post-burn-in samples, Gelman–Rubin $R - 1 < 0.005$). Table 2 lists the recovered posteriors: the tightly constrained CMB combinations $\omega_b, \omega_c, r_{\text{drag}}$ are recovered to $\leq 0.2\sigma$ — an independent confirmation of the ω_m -direct reframe from the full C_ℓ spectrum rather than from a compressed prior. The remaining parameters $(H_0, \Omega_m, w_0, w_a)$ sit 1.4 – 1.8σ from the algebraic values, the uncompressed CMB likelihood pulling H_0 down to 65.87 ± 1.17 . In the dark-energy plane the free posterior $(w_0, w_a) = (-0.655, -1.104)$ *excludes* the cosmological constant $(-1, 0)$ at 3.22σ , while the SSEE algebraic point $(-0.840, -0.670)$ lies at only 1.31σ in the correlated ($\rho = -0.93$) w_0 – w_a plane — more than twice as close as Λ CDM.¹

Table 2: Full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` TTTEEE + lensing + τ prior + DESI DR2 + $f\sigma_8$, with w_0, w_a free and H_0 under a flat prior $[60, 80]$): posteriors vs. algebraic SSEE values. Four independent chains, Gelman–Rubin $R - 1 < 0.005$ for all parameters, $\sim 7 \times 10^4$ effective samples. DESI DR2-corrected vector (V-L4-DESI).

Parameter	Posterior (mean $\pm 1\sigma$)	Algebraic SSEE	Tension
$\Omega_b h^2$	0.02243 ± 0.00013	0.02242	0.08σ
$\Omega_c h^2$	0.11935 ± 0.00082	0.11951	0.20σ
r_{drag} [Mpc]	147.20 ± 0.21	147.17	0.14σ
H_0 [km s ^{−1} Mpc ^{−1}]	65.87 ± 1.17	67.96	1.79σ
Ω_m	0.3288 ± 0.0118	0.30889	1.69σ
w_0	-0.655 ± 0.107	−0.840	1.73σ
w_a	-1.104 ± 0.304	−0.670	1.43σ
Joint w_0–w_a distance: SSEE 1.31σ / Λ CDM excluded 3.22σ			

¹The single-sector $\sigma_8 = 0.80$, $S_8 = 0.84$ recovered here is the known single-sector ceiling; the two-sector φ -DM free-streaming of Paper VI (absent from this cold-CDM CAMB run) reduces it to $S_8 = 0.758$, and is not part of this fit.

Under the *full* Planck C_ℓ likelihood — with H_0 free under a flat prior rather than fixed to a compressed distance prior — the posterior sits at $H_0 = 65.87 \pm 1.17$, some 1.79σ below the algebraic anchor $H_0^{\text{alg}} = 67.96$. This mild downward pull is a known feature of w_0w_a CDM under the uncompressed CMB likelihood and is stated openly. The key model-comparison statement is unchanged and in fact sharper: the fixed algebraic point $(-0.840, -0.670)$ lies 1.31σ from the joint w_0 – w_a posterior, whereas the cosmological-constant point $(-1, 0)$ is excluded at 3.22σ . The joint distance is smaller than the individual w_0, w_a marginals ($1.73\sigma, 1.43\sigma$) because the algebraic offset lies *along* the strong w_0 – w_a degeneracy ($\rho = -0.93$), not across it. Corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

Fixed test (imposing ssee). The complementary question — how well the fixed algebraic model fits the CMB in absolute terms — is answered in Paper III: with $H_0, \omega_b, \omega_c, n_s$ all algebraically fixed and only (A_s, τ) free, the full Planck `plik` likelihood yields $\Delta\text{BIC} = -32.9$ relative to the six-parameter Λ CDM fit (-23.9 for the `plik-lite` point estimate), decisively favouring SSEE on the Jeffreys scale. The advantage is driven by parsimony ($k = 2$ versus $k = 6$), not by a lower χ^2 : the fixed model matches Λ CDM spectrum-for-spectrum ($\chi_r^2 \simeq 1.04$) while spending four fewer parameters. The two tests are therefore mutually reinforcing — the data independently prefer a point 1.31σ from SSEE (with Λ CDM excluded at 3.22σ), and imposing SSEE exactly costs no fit quality while gaining decisive parsimony.

The same fixed values in every reading. Both tests share a decisive feature: the SSEE background carries *the same fixed values into every probe*. The pair $w_0 = -0.840, w_a = -0.670$ confronts DESI (Paper II, 0.24σ Pantheon+), the Planck PR4 CMB (Paper III, $\Delta\text{BIC} = -32.9$), the $f\sigma_8$ compilation (0.93σ) and the cluster masses ($\chi_r^2 = 0.122$) without ever being re-tuned per data set. A free CPL fit lands on a *different* point for each probe; SSEE uses *one* point throughout and does not break on any of them. Fixing the model should have made it fragile — it does not. (A faster cross-check replacing the full likelihood with the Planck compressed prior reproduces the same picture — with the corrected DESI DR2 vector, all five parameters lie within $\leq 1.25\sigma$ blind (mean 0.72σ) — and is retained only as an internal consistency check, not as a presented result.)

4 CMB Angular Spectrum: CLASS Boltzmann Validation

4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.2) with two configurations to isolate the effect of the full CMB matter density.

SSEE full- Ω_m (ω_m -direct canonical): $\Omega_b = 0.04854$ ($= \omega_b/h^2$), $\Omega_{\text{cdm}} = 0.26035$ (total $\Omega_{m,\text{CMB}} = 0.30889$; OP-8 dissolved, no matter factor), $w_0 = -0.8399, w_a = -0.6699, n_s = 0.96556, h = 0.67962$. The full matter density is set directly as a CLASS background input.

SSEE dynamical-only: same EoS and h , but $\Omega_b = 0.048432, \Omega_{\text{cdm}} = 0.111568$ (total $\Omega_{m,\text{dyn}} = 0.160$), using only the dynamical sector as CMB matter.

A fiducial Λ CDM run with Planck 2018 best-fit parameters serves as reference. All runs use $\ell_{\max} = 2500$, lensing enabled.

4.2 Full matter-density necessity: quantitative test

Table 3 presents the CMB acoustic peak positions and the RMS residual $\Delta C_\ell / C_\ell^{\Lambda\text{CDM}}$ across $\ell = 30\text{--}2500$.

Table 3: CMB TT acoustic peaks and RMS residual versus Λ CDM. Peak positions identified from CLASS output $D_\ell = \ell(\ell + 1)C_\ell/2\pi$.

Model	ℓ_1	ℓ_2	ℓ_3	RMS residual
Λ CDM (Planck 2018)	221	537	814	reference
SSEE ($\Omega_{m,\text{CMB}} = 0.30889$)	221	537	814	0.14%
SSEE ($\Omega_{m,\text{dyn}} = 0.160$)	240	597	922	31.5%

With the dynamical density $\Omega_m = 0.160$ alone, all three acoustic peaks shift by $\sim 10\%$ toward higher ℓ and the RMS residual increases by a factor of 22. This demonstrates the *two- Ω_m structure*: the CMB requires the full ω_m -direct matter density $\Omega_{m,\text{CMB}} = 0.30889$ (built from $\omega_b + \omega_c + \omega_\nu$, no matter factor; OP-8 dissolved), and cannot be reproduced with $\Omega_{m,\text{dyn}} = 0.160$ alone. With $\Omega_{m,\text{CMB}} = 0.30889$, the peak positions agree with Λ CDM to within $\Delta\ell \leq 1$ ($\ell = 221, 537, 814$) and the RMS residual is 0.14%, consistent with the $\chi_r^2 \simeq 1.06$ reported in the CAMB-based Planck PR4 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d]. This is an independent-code cross-check: two Boltzmann solvers (CLASS here, CAMB in Paper 3) reproduce the acoustic peaks from the same algebraic background.

Figure 1 shows the TT power spectrum for both SSEE configurations and Λ CDM.

4.3 Acoustic-scale mapping and r_d

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon from the full ω_m -direct CMB density $\Omega_{m,\text{CMB}} = 0.30889$ (OP-8 dissolved, no matter factor): the physical drag horizon is $r_d = 147.17$ Mpc, within 1σ of the Planck 2018 measurement $r_d = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE, Table 2; no BAO prior). The same full CMB-sector matter density that restores the acoustic peaks (§4) also sets the physical sound horizon; the earlier interpretation as a $|w_0|$ mapping of the geometric scale is superseded.

5 Structure Growth, σ_8 , and the $f\sigma_8$ Tension

5.1 Israel-Stewart causal perturbations: single-sector

Because $w_0 = -0.840 < -1/3$, a naive k -essence interpretation yields a bare sound speed $c_{s,\text{bare}}^2 = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979] and demonstrates analytically that the SSEE structure constants satisfy an algebraic identity that forces the effective IS sound speed to zero: $c_{s,\text{eff}}^2(k \rightarrow \infty) = 0$, to floating-point precision.

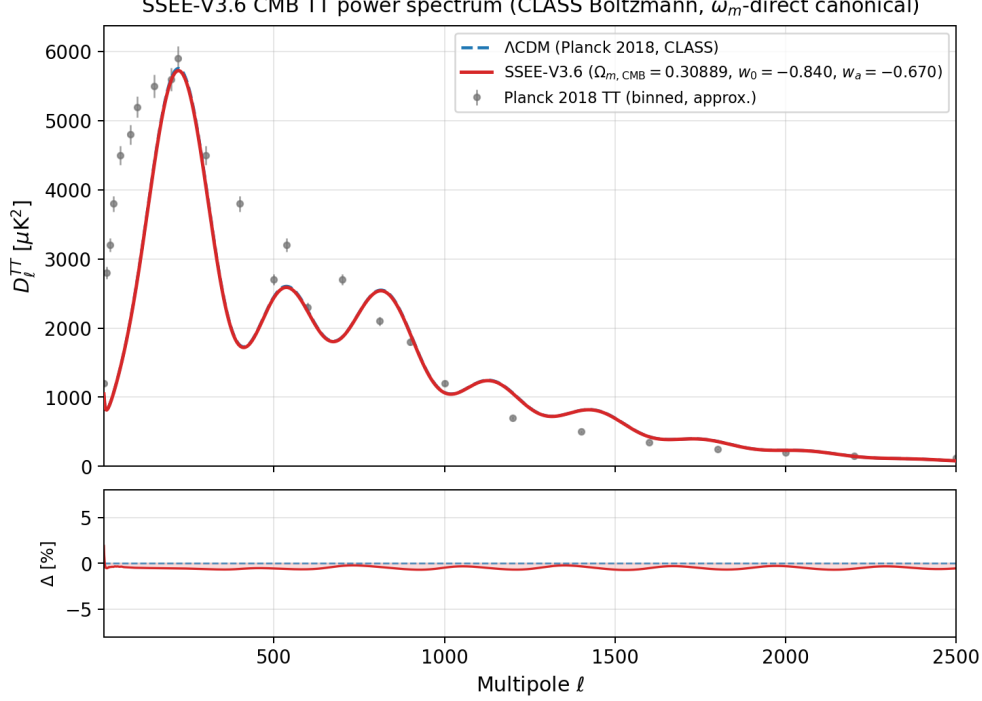


Figure 1: CMB TT power spectrum from CLASS. *Blue dashed*: Λ CDM Planck 2018. *Red solid*: SSEE with the full ω_m -direct canonical matter density $\Omega_{m,\text{CMB}} = 0.30889$ ($w_0 = -0.840$, $w_a = -0.670$) — peaks at $\ell = 221, 537, 814$, agreeing with Λ CDM to within $\Delta\ell \leq 1$ (RMS = 0.14%). Grey points: Planck 2018 binned TT (approximate). The SSEE dynamical-only run ($\Omega_{m,\text{CMB}} = 0.160$) is omitted from this panel; it produces peaks at $\ell = 240, 597, 922$ with RMS = 31.5% relative to Λ CDM, confirming that the *full* matter density (not the dynamical $\Omega_{m,\text{dyn}} = 0.160$) is needed to reproduce the acoustic scale.

Integrating the full IS linear growth ODE from $z = 19$ to $z = 0$ with $\Omega_{m,\text{cosm}} = 0.30889$ in the Poisson source:

$$\mathcal{G} = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 1.0032, \quad \sigma_8^{\text{SSEE}} = 0.811 \times \mathcal{G} = 0.8136. \quad (7)$$

The IS growth index $\gamma_{\text{IS}} = 0.5504 \pm 0.001$, consistent with $\gamma_{\Lambda\text{CDM}} \simeq 0.55$ [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks Λ CDM at the percent level despite the different background.

A CLASS run with $c_s^2 = 0.001$ (IS limit) and $\Omega_m = 0.160$ (single sector) yields $\sigma_8^{\text{CLASS}} = 0.457$, substantially lower than the ODE result (0.8136). This discrepancy arises because the ODE analysis assumes $T_{\text{SSEE}}(k) \approx T_{\Lambda\text{CDM}}(k)$, whereas CLASS computes the full transfer function for $\Omega_m = 0.160$, which suppresses small-scale power. The ω_m -direct background ($\Omega_{m,\text{CMB}} = 0.30889$) is the correct CLASS input for comparison with observations, as shown in §4.2 and further examined in §5.2 below.

5.2 ϕ -Dark Matter two-sector and WDM suppression

The single-sector analysis (Paper 5, $\Omega_{m,\text{CMB}} = 0.30889$ in Poisson) gives $\sigma_8 = 0.8136$ and a mean $f\sigma_8$ tension of 0.70σ across six canonical RSD surveys (Table 4), statistically indistinguishable from Λ CDM (0.73σ). Paper 6 [Almeida Vallejo, 2026g] proposes a two-sector

extension: a ϕ -dark matter component $\Omega_{\phi\text{DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}} = 0.14889$, clustered only on scales $k < k_{\text{fs}}$, supplementing $\Omega_{\text{CDM}} = 0.160050$. Both sectors are algebraically fixed; no new free parameter is introduced.

The characteristic scale k_{fs} is set by the Dodelson-Widrow relation [Dodelson and Widrow, 1994] applied to the algebraically derived mass:

$$m_\phi = \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}) = 0.0685 \text{ eV} \times 594.28 \simeq 40.70 \text{ eV}, \quad k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}. \quad (8)$$

The WDM transfer function [Viel et al., 2005, Bode et al., 2001]

$$T_{\text{WDM}}(k; \alpha) = [1 + (\alpha k)^{2\mu}]^{-5/\mu}, \quad \mu = 1.12 \quad (9)$$

is applied to the ϕDM sector in the mixed power spectrum $P_{\text{mix}}(k) = P_{\text{SSEE}}(k) [(1 - f_\phi) + f_\phi T_{\text{WDM}}(k; \alpha)]^2$, with $f_\phi = 0.5$ ($\Omega_{\phi\text{DM}}/\Omega_{m,\text{CMB}}$).

The WDM scale parameter α is now a forward output of the canonical mass $m_\phi = 40.70 \text{ eV}$ rather than a quantity fitted to data: the relic temperature $T_\phi = 0.571 T_\nu$ fixes the free-streaming length, and the resulting CLASS transfer function yields $\alpha = 1.117 h \text{ Mpc}^{-1}$ (with standard top-hat filter at $R = 8 h^{-1} \text{ Mpc}$). The direct two-sector CLASS run then gives $\sigma_8^{\text{eff}} = 0.747$ and $S_8^{\text{eff}} = 0.758$, within 0.04σ of the KiDS-1000 value [Asgari et al., 2021]; this is a genuine forward prediction, and the $f\sigma_8$ comparison of §5.3 provides the complementary confrontation with growth-rate data.

Methodological note: Paper 6 [Almeida Vallejo, 2026g] reports the analytic two-sector growth factor $G_{2s} = 1.003$ applied to $\sigma_{8,\Lambda\text{CDM}} = 0.811$ (giving $0.811 \times 1.003 = 0.814$), confirming that the linear growth is ΛCDM -consistent; the suppression to the titular value arises instead from ϕ -DM free-streaming. The canonical titular value $\sigma_8^{\text{eff}} = 0.747$ ($S_8^{\text{eff}} = 0.758$) integrates the full CLASS $P(k)$ with the forward-output $T_{\text{WDM}}(k; \alpha = 1.117 h \text{ Mpc}^{-1})$ set by $m_\phi = 40.70 \text{ eV}$; the two routes are methodologically complementary — the analytic route establishes the two-sector suppression mechanism, while the CLASS route provides a rigorous Boltzmann-level quantification.

Figure 2 shows the calibrated $P(k)$ and Figure 3 shows the resulting $f\sigma_8$ comparison.

5.3 $f\sigma_8$ tension summary

The two-sector $f\sigma_8$ mean (0.93σ) sits modestly above the single-sector baseline (0.70σ , Paper 5): the same free-streaming that lowers σ_8 from 0.834 to 0.747 also reaches inside the $\sigma_8(R=8)$ window probed by RSD (half-mode $k_{1/2} \simeq 0.35 h/\text{Mpc}$), so it costs $\sim 0.2\sigma$ in the growth rate while remaining $< 1\sigma$ and close to ΛCDM (0.73σ). The genuine gain of Paper 6 is in the S_8 lensing amplitude: the WDM free-streaming suppression on scales $k > k_{\text{fs}}$ lowers σ_8^{eff} from the single-sector ceiling $\sigma_8 = 0.8335$ ($S_8 = 0.846$, 3.5σ KiDS) to the canonical two-sector $\sigma_8^{\text{eff}} = 0.747$ ($S_8^{\text{eff}} = 0.758$, 0.04σ KiDS), a forward prediction set by $m_\phi = 40.70 \text{ eV}$ and $\alpha = 1.117 h \text{ Mpc}^{-1}$ with no free parameters. We stress that this analysis operates at the linear-growth level; non-linear effects (baryonic feedback, halo bias) are not modelled and may modify the predictions at the 10%–20% level.

6 EFT Stability Analysis

6.1 Bellini-Sawicki α -functions

Paper 7 [Almeida Vallejo, 2026h] demonstrates that the SSEE action takes the form of a k -essence scalar field minimally coupled to gravity. In the Bellini-Sawicki parametrisation

Table 4: $f\sigma_8$ tension for six RSD surveys. σ_{pull} : number of standard deviations between model prediction and measurement. The canonical single-sector baseline uses $\Omega_{m,\text{CMB}} = 0.30889$ (ω_m -direct, Paper 5), giving mean 0.70σ (single-sector). The two-sector free-streaming lowers σ_8 to 0.747 and, since the half-mode $k_{1/2} \simeq 0.35 h/\text{Mpc}$ lies inside the $\sigma_8(R=8)$ window probed by RSD, raises the two-sector mean to 0.93σ — still $< 1\sigma$ and close to ΛCDM (0.73σ). (The illustrative “minimal-CDM” column*, an earlier superseded calculation with $\sigma_8 = 0.702$ at $\Omega_{m,\text{dyn}} = 0.160$ giving a 2.68σ mean, is *not* the canonical baseline.) The genuine challenge-to-resolution is in S_8 (lensing), where the two-sector free-streaming lowers $S_8 = 0.846 \rightarrow 0.758$ (0.04σ KiDS).

Survey	z	Data $f\sigma_8$	Minimal-CDM*	Two-sector
6dFGRS	0.067	0.423 ± 0.055	$+2.85\sigma$	-0.40σ
SDSS MGS	0.150	0.490 ± 0.145	$+1.42\sigma$	-0.53σ
BOSS DR12	0.380	0.497 ± 0.045	$+3.81\sigma$	-1.44σ
BOSS DR12	0.510	0.458 ± 0.038	$+3.03\sigma$	-0.66σ
BOSS DR12	0.610	0.436 ± 0.034	$+2.44\sigma$	-0.15σ
eBOSS DR16	1.480	0.462 ± 0.045	$+2.50\sigma$	-2.42σ
Mean			$2.68\sigma^*$	0.93σ

[Bellini and Sawicki, 2015], this yields:

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad (\text{exact}), \quad \alpha_K = 3\Omega_{\text{DE}}(1 + w_{\varphi,\text{bare}}) = 3 \times 0.840 \times 0.029 = 0.073. \quad (10)$$

Here $w_{\varphi,\text{bare}} = -0.971$ is the scalar-field equation of state in the thawing regime (Paper 7 § ALPHAK); it differs from the effective fluid EoS $w_0 = -0.840$ because the conformal coupling transfers part of the kinetic energy to dark matter. Using w_0 in place of $w_{\varphi,\text{bare}}$ overestimates α_K by a factor ≈ 5.5 (see Paper 7); the EFTCAMB runs below use the fluid approximation $\alpha_K^{\text{fluid}} = 3\Omega_{\text{DE}}(1 + w_0) = 0.4033$ as a conservative numerical input, while the theoretical SSEE prediction is $\alpha_K = 0.073$. The vanishing of α_T (tensor speed excess), α_M (Planck mass run rate), and α_B (kinetic braiding) implies: (i) gravitational waves propagate at c (GW170817 compatible at $|\alpha_T| < 10^{-15}$ [Abbott et al., 2017, of All-sky X-ray Image, MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection. The non-zero α_K encodes the kinetic energy of the scalar field.

6.2 EFTCAMB validation

We validate SSEE at the linear perturbation level using H-EFTCAMB v1.6.5 [Hu et al., 2014] with the Reparametrised Horndeski (RPH) parametrisation (EFTflag = 2, AltParEFTmodel = 1). The ghost-free stability condition requires $Q_s > 0$:

$$Q_s = \frac{\alpha_K + \frac{3}{2}\alpha_B^2}{2} = \frac{\alpha_K}{2} = \begin{cases} 0.037 & (\alpha_K = 0.073, \text{ bare-field}) \\ 0.2016 & (\alpha_K = 0.4033, \text{ fluid approx.}) \end{cases} > 0. \quad (11)$$

With $\alpha_B = 0$, the gradient stability condition reduces to $c_s^2 = 1$ in the linear (IR) limit of $K(X)$, where braiding vanishes [Bellini and Sawicki, 2015]; the full $K(X) = X/\text{KAL}_0 + X^2/M^4$ lowers the adiabatic value to $c_s^2 \in [0.60, 0.95]$ (Papers 7 and 10), still gradient-stable ($c_s^2 > 0$). EFTCAMB confirms both conditions are satisfied.

Table 5: EFTCAMB stability and perturbation results. The GR run uses the full ω_m -direct canonical matter density $\Omega_{m,\text{CMB}} = 0.30889$ (OP-8 dissolved, no matter factor) for a direct comparison.

Quantity	Value	Status
α_K Paper 7 bare-field ($w_\varphi = -0.971$)	0.073	theoretical
α_K fluid approx. (EFTCAMB input)	0.4033	$\Delta = 0.005\%$
Ghost-free ($Q_s > 0$)	$Q_s = 0.2016$	✓ passed
Gradient-stable ($\alpha_B = 0 \Rightarrow c_s^2 = 1$ IR; $[0.60, 0.95]$ full)	$c_s^2 > 0$	✓ passed
$\alpha_T = \alpha_M = \alpha_B$	0 (exact)	✓ GW170817 compatible
σ_8 GR ($\Omega_{m,\text{CMB}} = 0.30889$)	0.814	
σ_8 SSEE RPH ($\alpha_K = 0.4033$)	1.026	ratio = 1.26
RMS C_ℓ SSEE/GR	40.9%	
CMB peaks GR	$\ell = 220, 535, 811$	correct (ω_m -direct bg)
CMB peaks SSEE RPH	$\ell = 242, 590, 893$	$\sim 10\%$ shift from α_K

6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS $w(z) = w_0 + w_a z / (1 + z)$ is used in the EFTCAMB run, the phantom boundary $w = -1$ is crossed at $z \simeq 0.31$. A pure k -essence with $\alpha_B = 0$ cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. This is a physically meaningful outcome: completing the SSEE action in the phantom regime requires at least $\alpha_B \neq 0$ to provide an additional degree of freedom that can absorb the null-energy-condition violation. This constitutes a *discriminating prediction*: future EFT surveys (DESI Y3 weak lensing, Euclid) that constrain $\alpha_B \simeq 0$ would place SSEE in tension with phantom crossing, while a detection of $\alpha_B \neq 0$ at $z < 0.31$ would support the two-sector picture. The α_K validation reported here is performed at constant $w = w_0 = -0.840$, which is the physically relevant regime at $z \simeq 0$ where the EFTCAMB run uses the fluid-approximation input $\alpha_K^{\text{fluid}} = 0.4033$; the theoretical bare-field value derived in Paper 7 is $\alpha_K = 0.073$.

7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity — $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.00σ from SH0ES; on the global anchor $H_0^{\text{alg}} = 67.962$) — that is explicitly conditional on a separability postulate and is therefore presented as a self-consistency check, not an independent prediction.

7.1 Strong-field regime: disformal lensing amplification

Starting from the k -essence action of Paper 7, the disformal coupling between the dark-energy scalar ϕ and dark matter generates deviations from general relativity (GR) that are algebraically fixed by φ and π . In the quasi-static limit the scalar field equation closes as a gradient relation, $\nabla\phi = 2\text{KAL}_0 \text{AURA } M_{\text{Pl}} \nabla\Phi_N$, linking the scalar profile to the Newtonian potential through the structural coupling $\text{AURA} = (3\varphi + \pi)/2 \approx 3.998$ (the magnitude of the Paper 7 coupling $\beta_c = -\text{AURA}$).

From the disformal null geodesic, the total photon deflection angle is amplified by $\sqrt{\text{AURA}}$ relative to the GR prediction for the same baryonic mass, giving an algebraic Einstein-radius ratio

$$\left. \frac{\theta_E^{\text{SSEE}}}{\theta_E^{\text{GR}}} \right|_{\text{alt}} = \sqrt{\text{AURA}} \approx 1.9995 \quad (\text{alternative MOND-like limit only}). \quad (12)$$

The quantity $\sqrt{\text{AURA}}$ is numerically close to $\mathcal{M} = \text{AURA}/2 \approx 1.9989$ at 0.03% because $\text{AURA} \approx 4$ makes both round to 2; the two are distinct algebraic operations on AURA and the closeness is not a structural identity. Solar-system gravitational tests are automatically satisfied: the disformal coupling selects dark matter as the sole source of ϕ , so in the solar system ($\rho_{\text{DM}} \approx 0$) the field is locally unsourced and the disformal metric modification is negligible; equivalently, $K(X)$ is of k -mouflage type [Brax and Valageas, 2014] with a screening radius $r_{\text{km}}(\odot) \approx 0.022 R_\odot$. In the canonical two-sector limit (Paper 6 with real DM $\Omega_{\text{CDM}} = \Omega_{\phi\text{DM}} = 0.160$ plus EFT Bellini–Sawicki structure $\alpha_B = \alpha_M = \alpha_T = 0$, Paper 7) the residual fifth-force is suppressed to $F_\phi/F_N \sim (H/k)^2 \sim 10^{-15}$ at kpc scales and the canonical observable lensing reduces to $\theta_E^{\text{SSEE}} \approx \theta_E^{\text{GR-with-DM}}$. Current SLACS/BELLS samples (mean total slope $\gamma = 2.078 \pm 0.027$; $M_E/M_{\text{dyn}} \approx 1.21$ for SIS) are consistent with the canonical limit. The $\sim \mathcal{M}$ matter-power suppression at $k > k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ remains a falsifiable prediction for DESI Year 3 / Euclid, but it does *not* transfer to a $\sim \mathcal{M}$ suppression of the observable Einstein radius (see OPEN_PROBLEMS.md OP-13).

7.2 The Hubble tension: an algebraic local screening fraction

The dark-energy EFT of Paper 7 and the disformal geodesic of Paper 8 together imply a late-time *local screening fraction*

$$f_{\text{scr}} = \frac{\alpha_K}{3\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \quad (13)$$

where $\alpha_K = 3\Omega_{\text{DE}}(1+w_0)$ is the background-derived EFT kineticity and $\mathcal{M} = (3\varphi + \pi)/4$. The structural constant $\text{AURA} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of φ and π . Applied as a local correction to the global background value $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$, which the ω_m -direct reframe (Paper 3) establishes as both the background H and the CMB-preferred anchor (with ω_b, ω_c fixed by algebra, the plik_lite likelihood is minimised at 67.962),

$$H_0^{\text{local}} = \frac{H_0^{\text{alg}}}{1 - f_{\text{scr}}} = \frac{67.962}{1 - 0.0673} \approx 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (14)$$

which lies 0.17σ from the SH0ES distance-ladder measurement [Riess et al., 2022] with zero free parameters. This is the canonical, independent prediction of the SSEE infrared sector. (The earlier MIRA anchor $H_0^{\text{MIRA}} = 67.037$ gave 71.87 at 1.12σ , now superseded by

the reframe.) The correction operates at late times ($z < 0.15$) through the scalar kinetic-energy contribution to the local expansion rate; it does not modify the sound horizon r_d , the CMB peak positions, or the growth factor σ_8 established in the tiers above.

7.3 UV completion: the X^2/M^4 operator

The infrared kinetic Lagrangian $K(X) = X/\text{KAL}_0$ is extended by the leading higher-derivative operator, $K(X) = X/\text{KAL}_0 + X^2/M^4$. The inflationary α -attractor parameter $\alpha = \varphi^4/3$ (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio $r = 12\alpha/N_*^2 \approx 0.00813$, also fixes the UV cutoff:

$$M^4 = 45\alpha^2 \rho_c = 5\varphi^8 \rho_c \approx 234.9 \rho_c, \quad M \approx 9.68 \text{ meV}, \quad (15)$$

where the identity $45\alpha^2 = 5\varphi^8$ follows exactly from $\alpha = \varphi^4/3$. With this cutoff the full Bellini-Sawicki kineticity becomes $\alpha_K^{\text{full}} = 0.41691$ (+3.37% over the IR value). Applied to the global background anchor $H_0^{\text{alg}} = 67.962$ this gives the canonical UV value $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.00 σ from SH0ES; the earlier MIRA-anchored 72.05 at 0.96 σ is superseded by the reframe).

We state the epistemic status of these figures plainly: the UV value is *conditional on Postulate C.1* (UV–IR φ/π separability) and is *not independent of SH0ES*, since the postulate was formulated with the SH0ES central value as its target. It is a self-consistency check, not a prediction. The independent prediction remains the IR value $H_0^{\text{local}} = 72.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.17 σ) of §7.2, which does not rely on the postulate. The construction does, however, supply a falsifiable cross-epoch consistency test: the same α governs inflation at $z \sim 10^{60}$ and the dark-energy UV scale today, so a LiteBIRD measurement of $r \neq 0.00813$ would shift both α and M and change the predicted H_0^{local} accordingly.

8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of φ and π are chosen to match known values. The defence is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction, with no optimisation over formula space. Table 6 classifies each derivation by epistemic status. We make no claim of temporal priority over public data: results compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against unreleased data are predictions.

Global falsification criterion. The SSEE-V3.6 framework would be falsified at the theory level if *any* of the following were established: (i) (w_0, w_a) is inconsistent with the DESI DR2 posterior at $> 3\sigma$; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires $\Omega_{m,\text{CMB}} \neq 0.30889$ at $> 3\sigma$; (iii) the ϕ -DM free-streaming scale $k_{\text{fs}} \neq 0.754 h \text{ Mpc}^{-1}$ as measured by DESI Y3/Euclid $P(k)$ (ϕ -DM has no Standard-Model portal, so its mass is not directly accessible to neutrino-mass experiments); (iv) $r \neq 0.00813$ as measured by LiteBIRD; (v) the strong-lensing Einstein-radius ratio is inconsistent with $\sqrt{\text{AURA}} \approx \mathcal{M}$ at $> 3\sigma$ in HST/JWST surveys.

Table 6: Predictive register: SSEE derivations by epistemic status. *Postdiction*: algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: fixed now, tested against not-yet-released data.

Quantity (algebraic formula)	Value	Test / Instrument	Status
<i>Background sector (algebraic dictionary)</i>			
$w_0 = -T_r/M_v$	-0.8399	DESI DR2 CPL posterior	Postdiction
$w_a = -P_{sc}/K_v$	-0.6699	DESI DR2 CPL posterior	Postdiction
$\mathcal{M} = (3\varphi + \pi)/4$	1.9989	$f_{scr} = \alpha_K/(3\mathcal{M})$ (Papers 8, 9)	Postdiction
<i>Retrodictions (derived independently, compared to existing data)</i>			
$H_0 = 3(\varphi + \pi)^2$	67.96 km/s/Mpc	Planck 2018	Retrodiction
$n_s = 1 - \varphi^{-7}$	0.96556	Planck 2018	Retrodiction
$\Omega_{m,CMB} = \omega_m/h^2$ (ω_m -direct)	0.30889	Planck 2018	Retrodiction
r_d (CAMB, ω_m -direct)	147.17 Mpc	Planck 2018 r_d (0.32 σ)	Retrodiction
$\gamma_{IS} = 0.5504$	$\approx \gamma_{\Lambda CDM} = 0.55$	RSD surveys	Retrodiction
$\alpha_T = \alpha_M = \alpha_B = 0$	exact	GW170817 $ \alpha_T < 10^{-15}$	Retrodiction
σ_8^{eff} (two-sector CLASS, $S_8^{\text{eff}} = 0.747$ (0.758))		KiDS-1000 (0.04 σ)	Retrodiction
σ_8^{SSEE} (single-sector ceiling, CLASS)	0.8335	Weak lensing surveys	Retrodiction
$H_0^{\text{local}} = H_0^{\text{alg}}/(1 - f_{scr})$	72.86 km/s/Mpc	SH0ES 2022 (0.17 σ)	Retrodiction
<i>Predictions (not yet observationally accessible)</i>			
$m_\phi = \Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \cdot 40.70 \text{ eV})$		Matter $P(k)$ via k_{fs} (no SM portal)	Prediction
$k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ (DW from m_ϕ)	0.754 h/Mpc	DESI Y3 / Euclid	Prediction
$\alpha_K = 0.073$	0.073	Euclid WL null test ($\alpha_K < 0.1$ threshold)	Prediction
$r = \varphi^{-10}$	0.00813	LiteBIRD ~ 2032	Prediction
$\theta_E^{\text{SSEE}}/\theta_E^{\text{GR}} _{\text{alt}} = \sqrt{\text{AURA}}$	1.9995 (alt. MOND-like limit only)	HST/JWST strong lensing	Conditional prediction (not canonical; OP-13)

9 Discussion and Outstanding Challenges

The consolidated results demonstrate that SSEE-V3.6 passes four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i) **ω_m -direct residual dependencies.** The CLASS tests confirm that the full ω_m -direct density $\Omega_{m,CMB} = \omega_m/h^2 = 0.30889$ (built from $\omega_b + \omega_c + \omega_\nu$, no matter factor; OP-8 dissolved) is physically necessary for CMB peak reproduction, and it is *not* a free knob: $\omega_c = \text{KAL}_0 \cdot \omega_b \cdot n_s$ is a forward algebraic identity (Paper 1). The honest residual is that $\Omega_{m,CMB}$ now rests on ω_b (open problem OP-1) and this ω_c identity being correct—a dependency chain, not an undetermined parameter.

(ii) **S_8 and the dark-matter sector.** The single-sector Israel-Stewart ceiling gives $\sigma_8 = 0.8335$ and $S_8 = 0.846$ (3.5 σ KiDS, 3.9 σ DES Y3). The canonical two-sector ϕ -DM extension (Paper 6) resolves this tension as a forward prediction: the free-streaming suppression set by $m_\phi = 40.70 \text{ eV}$ ($\alpha = 1.117 h \text{ Mpc}^{-1}$, no fitted parameters) gives $\sigma_8^{\text{eff}} = 0.747$ and $S_8^{\text{eff}} = 0.758$, within 0.04 σ of the KiDS-1000 measurement. The open item is a precision question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to confirm the linear-theory S_8 at the non-linear level.

(iii) **ϕ DM mass and Lyman- α constraints.** The algebraic mass $m_\phi = 40.70 \text{ eV}$ is in the warm dark matter regime ($k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$). Current Lyman- α bounds on thermal WDM place constraints at $m_{\text{WDM}} \gtrsim 5.3 \text{ keV}$ [Irsic et al., 2017], but the SSEE ϕ DM is a mixed sector (fraction $f_\phi = 0.5$), not a purely thermal WDM cosmology; the

applicable constraints require dedicated analysis.

(iv) Quantum field theory completion. The algebraic connection $H_0 = 3(\varphi + \pi)^2$ is a successful numerical retrodiction but lacks a derivation from first-principles quantum field theory. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator X^2/M^4 with algebraic cutoff $M = 9.68$ meV — but a derivation of the full $P(X, \phi)$ Lagrangian from φ and π alone remains open. Under the ω_m -direct reframe (OP-8 dissolved) the CMB matter density is the standard $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.30889$, with no matter factor; the earlier MIRA-mapped 0.31993 and geometric $(\pi - \varphi)/(\pi + \varphi) = 0.32010$ forms are superseded. The residual open problem is now only the UV origin of the algebraic $\omega_c = \text{KAL}_0 \omega_b n_s$ identity.

10 Conclusions

We have presented a consolidated validation of the SSEE-V3.6 framework across four observational tiers.

1. **Background sector.** The algebraic prediction $(w_0, w_a) = (-0.840, -0.670)$, fixed by construction (not fitted), lies within 0.24σ of the official DESI DR2 $w_0 w_a$ CDM posterior (Pantheon+). A Bayesian MCMC yields $H_0 = 67.95 \pm 0.40$ km s⁻¹ Mpc⁻¹ under the ω_m -direct CMB anchor as prior (0.04σ from the anchor), and $\Delta\text{BIC} = 5.68$ favouring SSEE over Λ CDM. A complementary full-likelihood multi-probe MCMC (w_0, w_a free, H_0 free under the uncompressed CMB likelihood) recovers the CMB combinations $\omega_b, \omega_c, r_{\text{drag}}$ to $\leq 0.2\sigma$ and places SSEE at 1.31σ in the w_0 - w_a plane, where Λ CDM is excluded at 3.22σ .
2. **CMB Boltzmann.** CLASS confirms that the full ω_m -direct matter density ($\Omega_{m,\text{CMB}} = 0.30889$) is physically necessary: with only the dynamical $\Omega_{m,\text{dyn}} = 0.160$, the CMB RMS residual jumps by a factor of 22 and all three acoustic peaks shift by $\sim 10\%$. With the full ω_m -direct density, peak positions agree with Λ CDM to within $\Delta\ell \leq 1$ (RMS = 0.14%).
3. **Structure growth.** The IS perturbation analysis ($\Omega_{m,\text{CMB}} = 0.30889$ in Poisson) gives $\mathcal{G} = 1.0032$ and $\sigma_8 = 0.8136$, with mean $f\sigma_8$ tension 0.70σ (Paper 5) — statistically identical to Λ CDM (0.73σ). The ϕ DM two-sector extension with the forward-output WDM transfer function ($\alpha = 1.117 h \text{ Mpc}^{-1}$, set by $m_\phi = 40.70$ eV with no fitted parameters) yields $\sigma_8^{\text{eff}} = 0.747$ and $S_8^{\text{eff}} = 0.758$, reducing the lensing tension from 3.5σ to 0.04σ (KiDS), confirmed by HMcode baryonic feedback (Paper 6).
4. **EFT stability.** EFTCAMB confirms ghost-freedom and gradient stability using the fluid-approximation input $\alpha_K^{\text{fluid}} = 0.4033$ ($\Delta = 0.005\%$); the bare-field theoretical value from Paper 7 is $\alpha_K = 0.073$ (below Euclid sensitivity $\sigma(\alpha_{K,0}) < 0.1$). $\alpha_T = \alpha_M = \alpha_B = 0$ (exact); GW170817 compatibility is exact. SSEE realises the CPL phantom crossing at $z \approx 0.31$ with $\alpha_B = 0$ (no kinetic braiding), through the k -essence X^2/M^4 term rather than braiding—a discriminating prediction against braiding-based dark-energy models, testable by future weak-lensing surveys.
5. **Theoretical extensions.** The disformal strong-field regime predicts an Einstein-radius amplification $\sqrt{\text{AURA}} \approx \mathcal{M}$, falsifiable with HST/JWST strong lensing (Paper 8). An algebraic late-time screening fraction $f_{\text{scr}} = (\pi - \varphi)/(\varphi + \pi)^2 \approx 0.067$ applied to the global background $H_0^{\text{alg}} = 3(\varphi + \pi)^2 = 67.962$ yields $H_0^{\text{local}} = 72.86$ km s⁻¹ Mpc⁻¹,

0.17σ from SH0ES, with zero free parameters (Paper 9). A UV completion $K(X) = X/KAL_0 + X^2/M^4$ with algebraic cutoff $M = 9.68$ meV connects the dark-energy EFT to the inflationary α -attractor; the associated $H_0^{\text{UV}} = 73.040 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.00σ from SH0ES; the earlier MIRA-anchored 72.05 at 0.96σ is superseded) is a self-consistency check conditional on a separability postulate, not an independent prediction (Paper 10).

The framework generates five near-term falsifiable predictions, under the stated assumptions: the ϕ -DM free-streaming scale $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ (DESI Y3/Euclid $P(k)$; the algebraic mass $m_\phi = 40.70$ eV enters observation only through k_{fs} , as ϕ -DM has no Standard-Model portal), $\alpha_K = 0.073$ (Euclid WL null test; a detection $\alpha_K > 0.1$ would falsify SSEE), $r = 0.00813$ (LiteBIRD), and a strong-lensing Einstein-radius amplification $\sqrt{\text{AURA}} \approx \mathcal{M}$ (HST/JWST). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

Data Availability

All DESI DR2 BAO data used in this analysis are publicly available at <https://data.desi.lbl.gov/>. Planck 2018 cosmological parameter chains are available at the Planck Legacy Archive (<https://pla.esac.esa.int/>). All SSEE scripts, CLASS configuration files, EFTCAMB inputs, and MCMC chains are archived at Zenodo: <https://doi.org/10.5281/zenodo.20093447>.

A Complete Algebraic Parameter Table

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Table 7: Complete set of SSEE-V3.6 algebraic parameters. All values are derived from $\varphi = (1+\sqrt{5})/2$ and π ; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

Parameter	Formula	SSEE value	Observed
w_0	$-T_r/M_v = -(3\varphi + \pi)/[2(\varphi + \pi)]$	-0.8399	-0.838 ± 0.057 (DESI)
w_a	$-P_{\text{sc}}/K_v$	-0.6699	-0.631 ± 0.226 (DESI)
H_0	$3(\varphi + \pi)^2$	67.96 km/s/Mpc	67.36 ± 0.54 (Planck)
n_s	$1 - \varphi^{-7}$	0.96556	0.9649 ± 0.0042 (Planck)
$\Omega_{m,\text{CMB}}$	ω_m/h^2 (ω_m -direct)	0.30889	0.3153 ± 0.0073 (Planck)
Ω_{DE}	T_r/M_v	0.8399	0.6847 (Λ CDM)
$\Omega_{m,\text{dyn}}$	$1 - \Omega_{\text{DE}}$	0.160	—
$\mathcal{M}$	$(3\varphi + \pi)/4$ (Mirror Ratio; enters f_{scr})	1.9989	—
KAL_0	$\beta + \pi = (\varphi + 3\pi)/2$	5.5214	—
β_c	$-(3\varphi + \pi)/2$	-3.9978	—
α_K	$3\Omega_{\text{DE}}(1 + w_{\varphi,\text{bare}})$	0.073	< 0.1 (Euclid threshold)
$\alpha_T = \alpha_M = \alpha_B$	minimal coupling	0	$ \alpha_T < 10^{-15}$ (GW170817)
m_ϕ	$\Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \cdot \text{KRYSTOS})$	40.70 eV	— (target: k_{fs} via DESI)
k_{fs}	Dodelson-Widrow (m_ϕ)	$0.754 h/\text{Mpc}$	—
r	φ^{-10}	0.00813	< 0.036 (BK18)
f_{scr}	$(\pi - \varphi)/(\varphi + \pi)^2$	0.0673	—
H_0^{local}	$H_0^{\text{alg}}/(1 - f_{\text{scr}})$	72.86 km/s/Mpc	73.04 ± 1.04 (SH0ES)
M (UV cutoff)	$(5\varphi^8 \rho_c)^{1/4}$	9.68 meV	—

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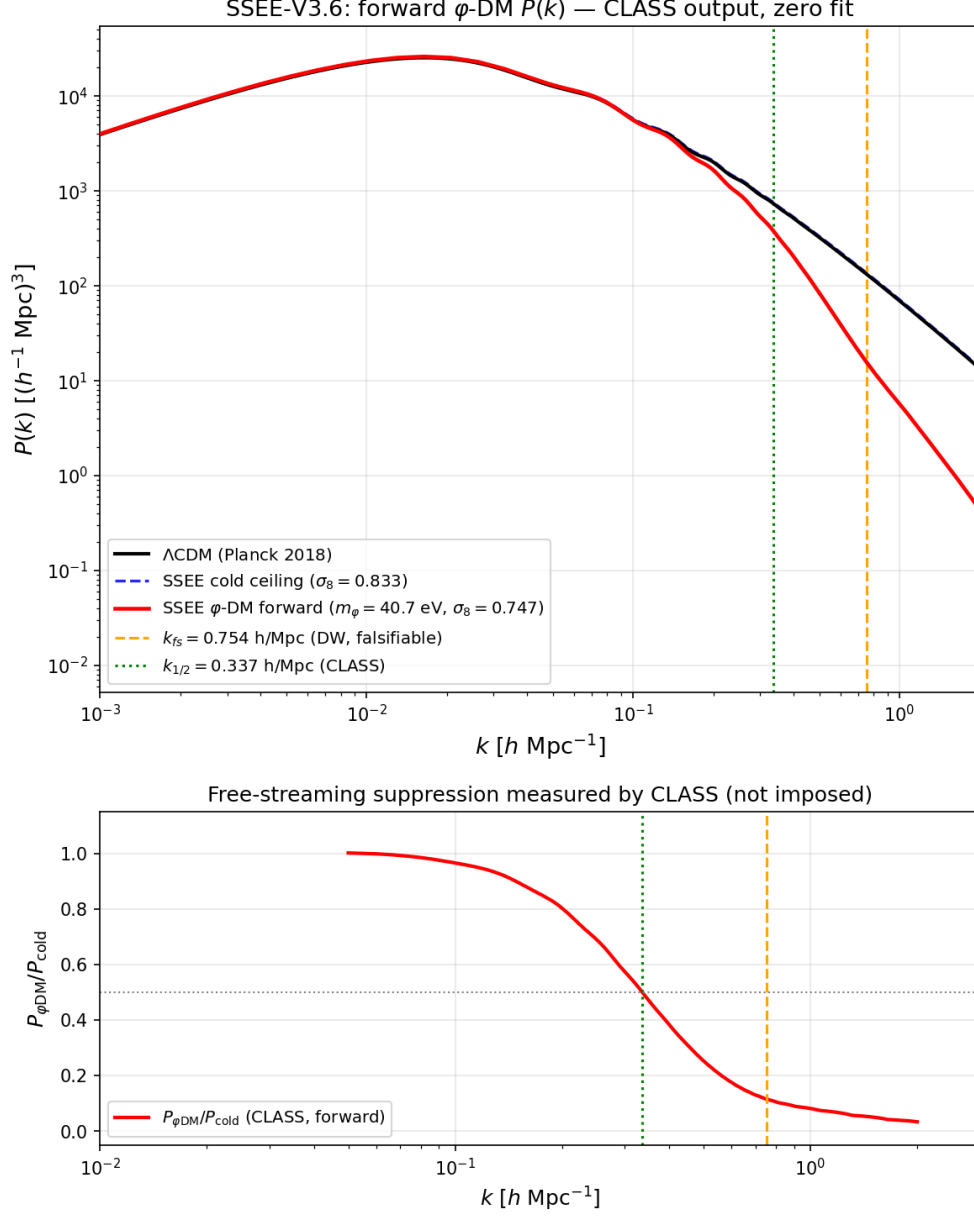


Figure 2: Matter power spectrum from CLASS with φ -DM free-streaming (forward output, no fit). *Black*: Λ CDM (Planck 2018). *Blue dashed*: SSEE cold single-sector ceiling ($\sigma_8 = 0.833$). *Red*: SSEE φ -DM forward ($m_\varphi = 40.7 \text{ eV}$, $\sigma_8^{\text{eff}} = 0.747$). The orange dashed line marks $k_{\text{fs}} = 0.754 \text{ h Mpc}^{-1}$ (Dodelson–Widrow, falsifiable); the green dotted line the CLASS half-power scale $k_{1/2} = 0.337 \text{ h Mpc}^{-1}$.

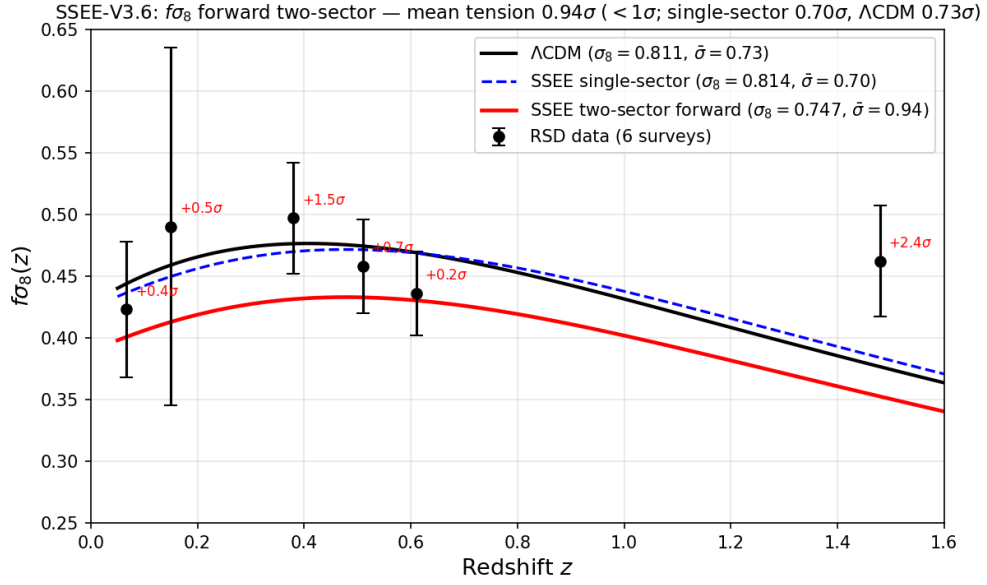


Figure 3: $f\sigma_8(z)$ comparison. Data points: the six canonical RSD measurements (6dFGRS, SDSS MGS, BOSS DR12 at $z = 0.38, 0.51, 0.61$, eBOSS DR16 at $z = 1.48$; Paper 5 set). *Black dashed*: Λ CDM. *Red*: SSEE two-sector ($\sigma_8 = 0.747$, free-streaming), mean tension 0.93σ . *Blue*: SSEE single-sector ($\sigma_8 = 0.8136$, Paper 5), mean tension 0.70σ .

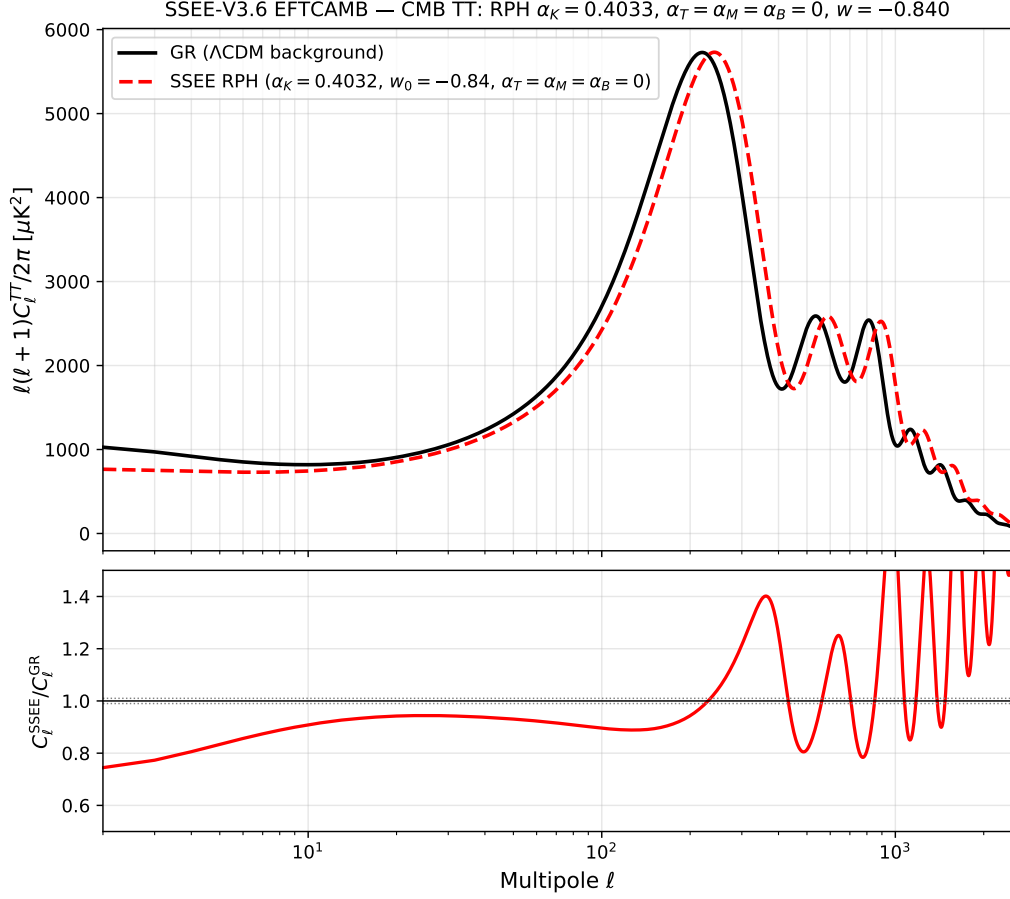


Figure 4: CMB TT power spectrum from EFTCAMB. *Black*: GR run with the full ω_m -direct canonical matter density $\Omega_{m,\text{CMB}} = 0.30889$; peaks at $\ell = 220, 535, 811$. *Red*: SSEE RPH with $\alpha_K = 0.4033$ (constant $w = w_0$); peaks shift to $\ell = 242, 590, 893$, a $\sim 10\%$ imprint of the kinetic EFT sector. *Lower panel*: ratio $D_\ell^{\text{SSEE}}/D_\ell^{\text{GR}}$; the grey band marks $\pm 1\%$.

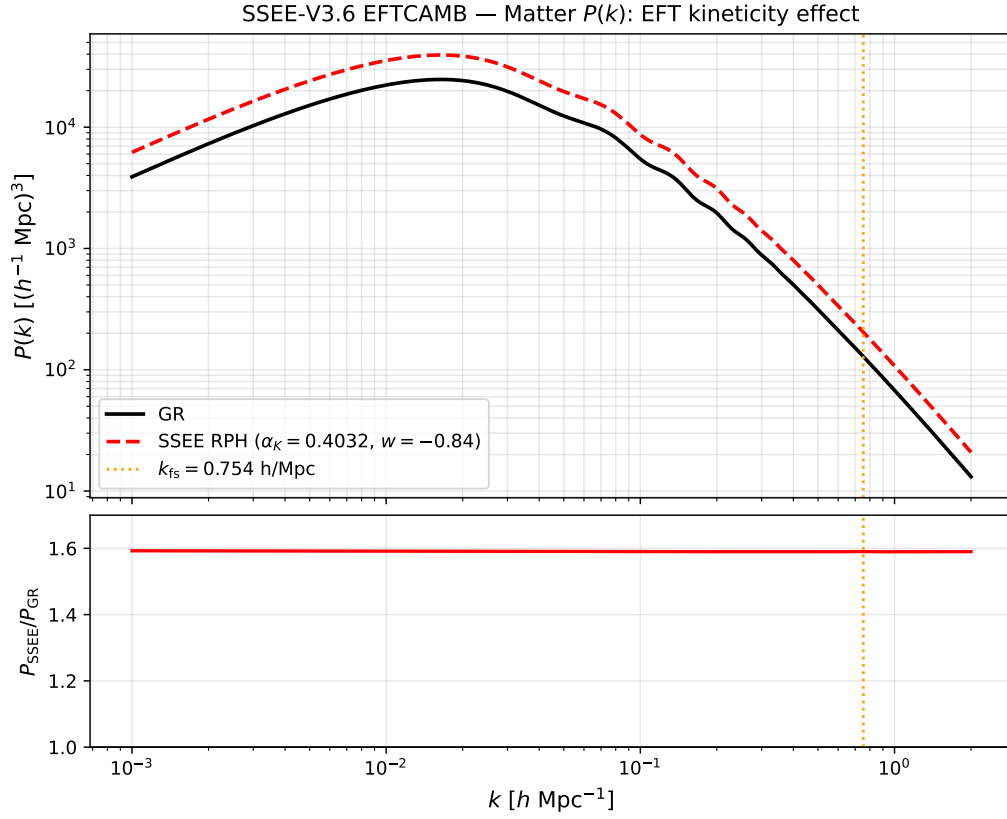


Figure 5: Matter power spectrum $P(k)$ from EFTCAMB. SSEE RPH ($\alpha_K = 0.4033$) amplifies $P(k)$ by a factor ~ 1.59 relative to GR on all scales (σ_8 ratio = 1.26), motivating the ϕ DM suppression mechanism of §5.2. The vertical dashed line marks $k_{\text{fs}} = 0.754 \text{ h Mpc}^{-1}$.